

Plan Change — System Design (HLD & LLD)

Item	Value
System	Plan Change
API	Plan Change API v1.0.0
Primary capability	Subscriber changes mobile plan online

This document describes the **Plan Change system** — authentication, catalog, order, payment, and activation.

Part 1 — High-Level Design (HLD)

1.1 Purpose

Allow a mobile subscriber to **change their current plan**: sign in, compare current and recommended plans, place a switch order, pay, and activate the new plan.

1.2 Actors

Actor	Role
Subscriber	Uses mobile app / web to change plan
Plan Change API	Backend REST services for the journey
Payment provider	Processes payment (UPI / card / wallet)
QA	Executes automated and manual tests

1.3 Scope

In scope (plan-change journey)

Step	Capability	Summary
1	Login	Authenticate with MSISDN + password; optional token refresh
2	PlanFetch	List current and recommended plans
3	PlanSwitch	Create plan switch order (order_id)
4	Payment	Pay for order and activate plan

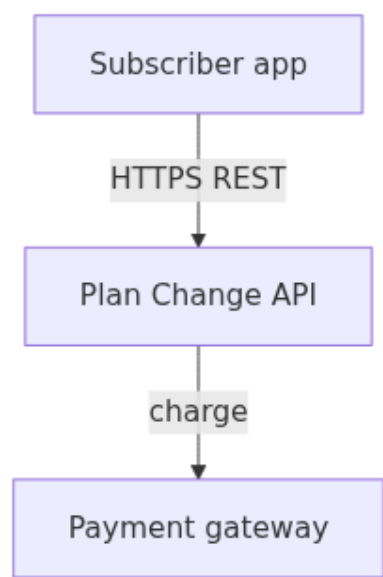
Also specified (not in default v1 journey)

Capability	Role
EligibilityCheck	Validate eligibility before switch (v2 extension)
SupportTicket, ProfileView, VASOverview	Separate product areas — share login only

Out of default journey

Support, profile, and VAS APIs exist in the system but are not part of the core plan-change flow.

1.4 Context diagram



1.5 Journeys by release (v1, v2, v3)

Version	Change vs previous
v1	Baseline plan-change journey
v2	Adds EligibilityCheck before switch
v3	Replaces direct Payment with OTP payment chain

Journey — v1 (baseline)



Flow: Login → PlanFetch → PlanSwitch → Payment

Journey — v2 (eligibility)



Flow: Login → PlanFetch → **EligibilityCheck** → PlanSwitch → Payment

Journey — v3 (OTP payment)

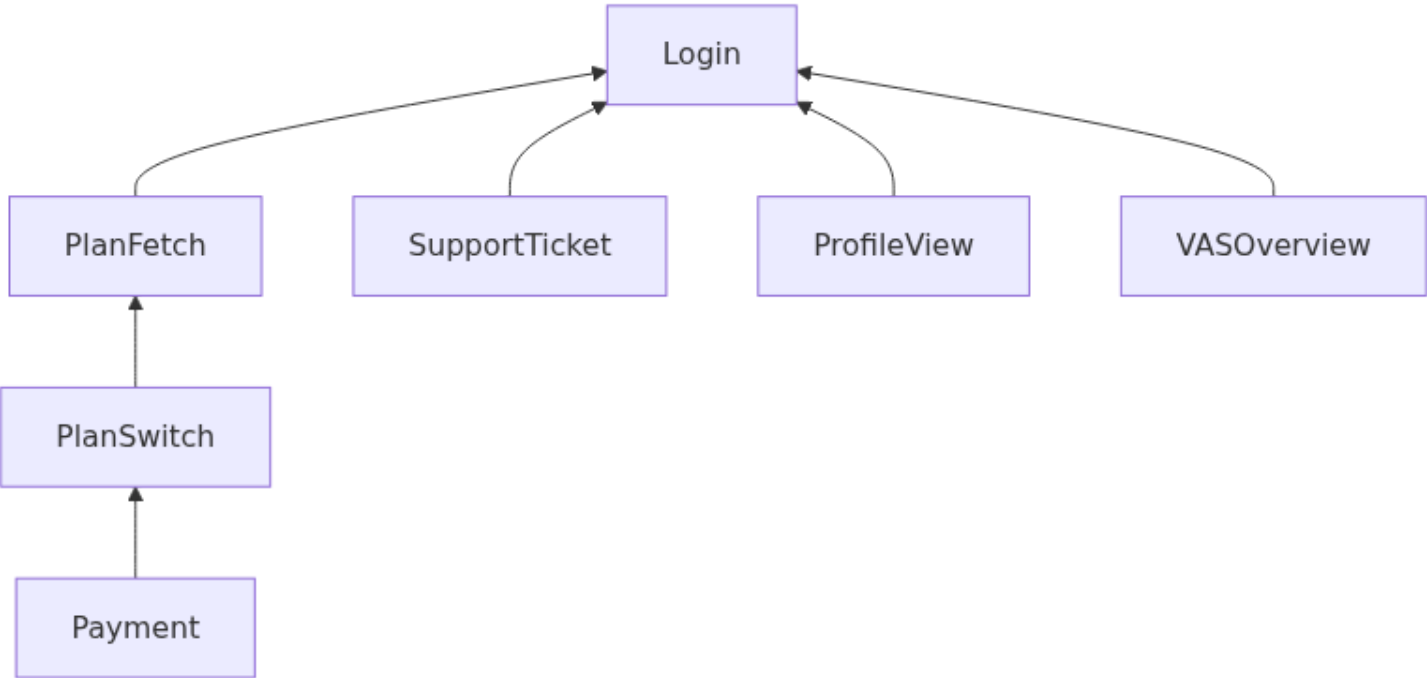


Flow: Login → PlanFetch → PlanSwitch → **PaymentInitiate** → **PaymentVerifyOtp** → **PlanActivate**

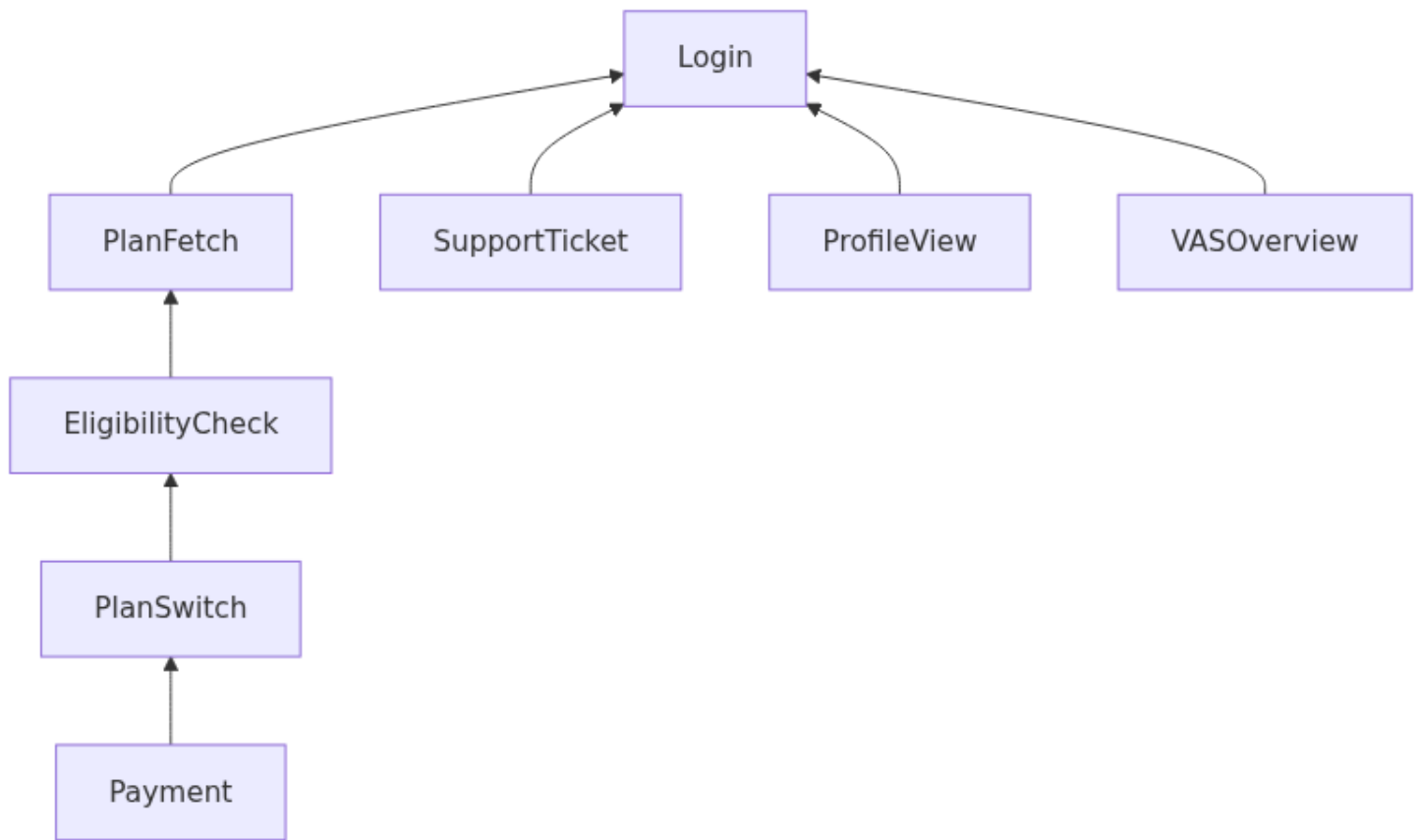
1.6 Capability dependencies by release

Arrows: dependent capability → prerequisite. Standalone capabilities (Support, Profile, VAS) depend only on **Login**.

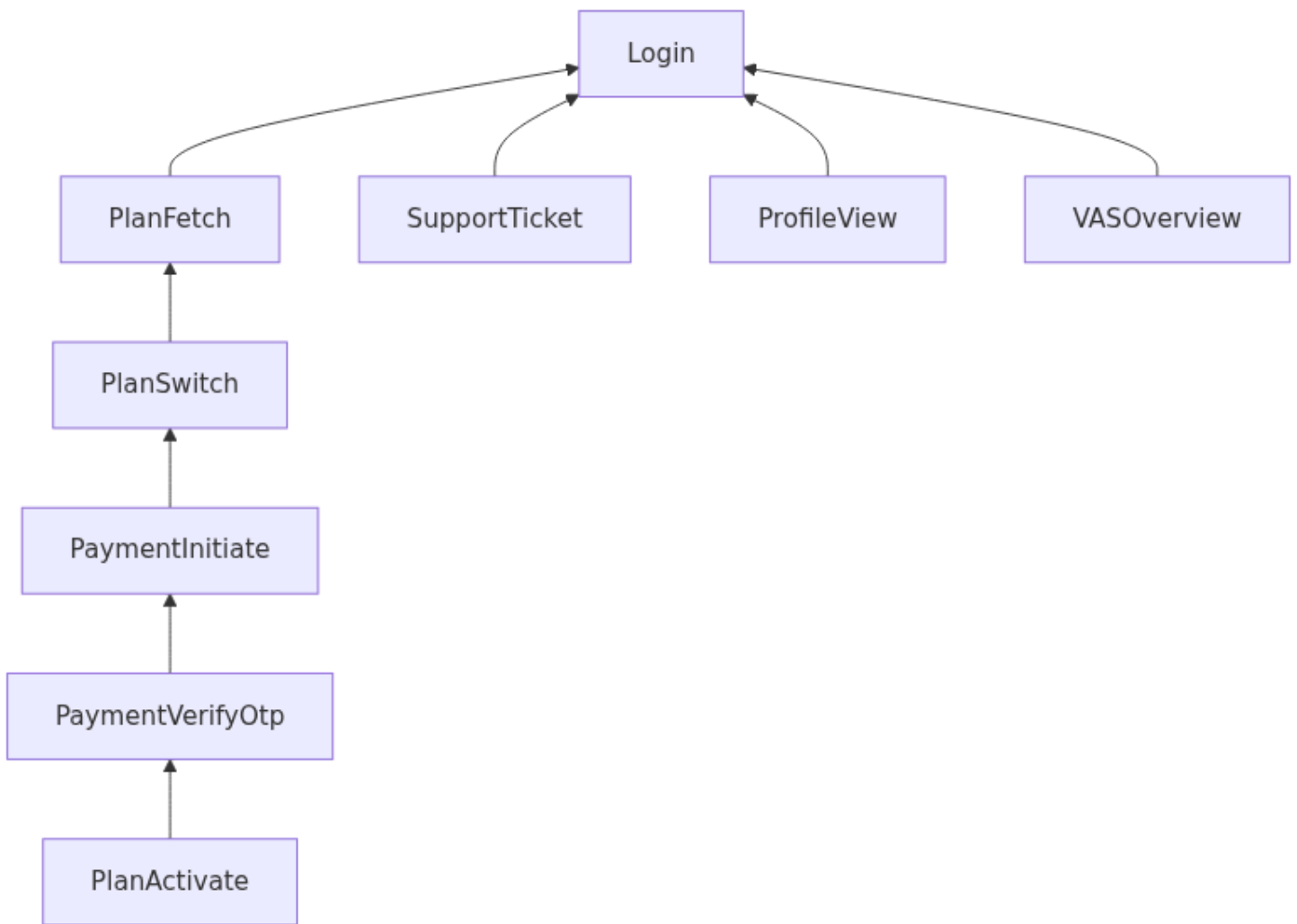
Capabilities — v1



Capabilities — v2



Capabilities — v3



v3 drops **EligibilityCheck** from the journey and replaces **Payment** with the OTP payment chain.

1.8 API landscape

Domain	Endpoints	Purpose
Authentic ation	POST /auth/login, POST /auth/token/refresh	Session
Plans	GET /plans, POST /plans/switch, POST /plans/eligibility-check, POST /plans/activate	Catalog, order, eligibility, activation (v3)
Payments	POST /payments/pay, POST /payments/activate (v1/v2); POST /payments/initiate, POST /payments/verify-otp (v3)	Pay and activate
Support	POST /support/tickets	Tickets (standalone)
Profile	GET /profile/details	Profile (standalone)
VAS	GET /vas/subscriptions	Subscriptions (standalone)

1.9 Non-functional requirements

Area	Requirement
Security	Authenticated access after login; 401 when session invalid
Data	Operations keyed by msisdn, order_id, plan_id
Errors	Documented 4xx/402 responses for negative scenarios
Testability	Positive and negative tests per critical capability

Part 2 — Low-Level Design (LLD)

2.1 Requirement specification

Field	Specification
Title	Plan Change
Description	Subscriber changes mobile plan: login → view plans → switch → pay → activate
Flow order	Login, PlanFetch, PlanSwitch, Payment
APIs referenced	POST /auth/login, GET /plans, POST /plans/switch, POST /payments/pay, POST /payments/activate

2.2 Capability specifications

Capability	Description	APIs	Depends on
Login	Authenticates user; returns session token	/auth/login, /auth/token/refresh	—
PlanFetch	Current and recommended plans	GET /plans	Login
EligibilityCheck	Validates plan eligibility (v2)	POST /plans/eligibility-check	PlanFetch
PlanSwitch	Creates switch order	POST /plans/switch	EligibilityCheck (v2); PlanFetch (v1, v3)
Payment	Pay and activate (v1/v2)	POST /payments/pay, POST /payments/activate	PlanSwitch
PaymentInitiate	Starts OTP payment (v3)	POST /payments/initiate	PlanSwitch
PaymentVerifyOtp	Verifies payment OTP (v3)	POST /payments/verify-otp	PaymentInitiate
PlanActivate	Activates plan after OTP (v3)	POST /plans/activate	PaymentVerifyOtp
SupportTicket	Raise support ticket	POST /support/tickets	Login
ProfileView	Read profile	GET /profile/details	Login
VASOverview	List VAS	GET /vas/subscriptions	Login

Authentication

Operation	Method	Path	Request	Success	Error
Login	POST	/auth/login	msisdn, password	200, token, user_id	401
Refresh token	POST	/auth/token/refresh	refresh_token	200	—

Plans

Operation	Method	Path	Request	Success	Error
List plans	GET	/plans	Query: type (current or recommended), msisdn	200, plans[]	401
Switch plan	POST	/plans/switch	msisdn, plan_id	200, order_id	400
Eligibility	POST	/plans/eligibility-check	msisdn, plan_id	200, eligible, reason	400

Payments

Operation	Method	Path	Request	Success	Error
Pay	POST	/payments/pay	order_id, payment_method (upi, card, or wallet), amount	200	402
Activate (after pay)	POST	/payments/activate	order_id	200	404
Initiate OTP payment (v3)	POST	/payments/initiate	order_id, payment_method	200, OTP sent	—
Verify OTP (v3)	POST	/payments/verify-otp	order_id, otp	200	400
Activate plan (v3)	POST	/plans/activate	order_id	200	404

Standalone APIs

Operation	Method	Path
Support ticket	POST	/support/tickets
Profile	GET	/profile/details
VAS list	GET	/vas/subscriptions

2.4 Test specification

Test ID	Scope	Type	Title	Expected
TC-login-001	Login	Positive	Valid login with correct credentials	HTTP 200, session token
TC-login-002	Login	Negative	Login with wrong password	HTTP 401
TC-API-login-401	POST /auth/login	Negative	API returns 401 for invalid credentials	HTTP 401
TC-planfetch-001	PlanFetch	Positive	Fetch current and recommended plans	HTTP 200
TC-planfetch-002	PlanFetch	Negative	Fetch plans without authentication	HTTP 401

Test attributes

Attribute	Values
type	positive, negative
test_layer	api
steps	Ordered actions
expected_result	HTTP status and outcome

2.5 Data model (logical)

Entity	Attributes	Used in
Subscriber	msisdn	Login, plans, switch, eligibility
Session	token, user_id	Authenticated calls
Plan	plan_id	Switch, eligibility
Order	order_id	Payment, activation
Payment	payment_method, amount	Pay and activate

2.6 Sequence — happy path (v1)

